

Muhammad Abdullah

Structural Engineer

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EDUCATION

University of Illinois Urbana-Champaign

MS Civil Engineering (Structural)

Coursework: Advanced Steel Design, Structural Dynamics, Finite Element Analysis

Aug. 2024 – May 2026

CGPA: 3.93

Military College of Engineering, NUST

BE Civil Engineering

Oct. 2017 – Jul. 2021

CGPA: 3.82

EXPERIENCE

Graduate Researcher

Smart Structures Technology Lab, University of Illinois

Mar. 2025 – May 2026

Urbana, IL, US

- Integrated reliability constraints into a legacy stochastic topology optimization framework
- Implemented and validated adjoint sensitivity formulations to compute gradients for highly non-linear reliability constraints
- Refactored core MATLAB scripts into a modular architecture to allow for greater flexibility in adding components to the framework

Management Trainee Engineer

National Logistics Cell

Feb. 2022 – Mar. 2023

D.I. Khan

- Managed execution of structural and road components in highway projects and inspection of bridge elements
- Responsible for supervising sub-contractors, and coordinating between consultant and contractor staff
- Prepared daily work reports, presentations, and updated monthly work schedules and progress
- Prepared bar bending schedules for building construction projects, and coordinated procurement
- Prepared a hydraulic report proposing an alternate design of a large culvert which was accepted. Alternate design resulted in significant time and cost savings for the contractor.
- Created an Android app linked to Google Sheets to track rebar inventory and issue details for multiple subcontractors, providing real-time summaries of currently held and issued rebar by type.

Intern

Designmen Consulting

Nov. 2021 – Jan. 2022

Islamabad

- Gained familiarity with high-rise residential tower drawings/plans, and assisted with inspection of site-work

PROJECTS

2D Frame Analysis Program using Python

- Created program capable of calculating the reactions and deformations of a 2D frame structure via an implementation of the Direct Stiffness Method. (Github)

Optimization of Viscous Damper Distribution in a 3D Frame Structure

- Developed a MATLAB-based topology optimization framework to determine the optimal distribution of viscous dampers in a 3D asymmetric frame structure under lateral drift and rotation objectives
- Extracted and processed structural matrices from SAP2000 to rebuild the model inside MATLAB, creating a more effective pipeline while testing different variations in the base structure

AWARDS

Fulbright Foreign Student Scholarship

U.S. Department of State & USEFP

Aug. 2024 – May 2026

- Received a fully funded scholarship for an MS program in Civil Engineering at the University of Illinois Urbana-Champaign

TECHNICAL SKILLS

Software: ETABS, SAP2000, Tekla Structures, STAAD.Pro, Excel, Revit, AutoCAD, Primavera, QGIS, HEC-RAS

Computational: Python, MATLAB, Grasshopper